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Joseph Henry Briggs

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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Introduction Chapter (??): Not started. Will write last/almost last
- IFO Chapter (??): Apparently this is a good chapter...
- PSL Chapter (??): Passed first drafting. Very Minor corrections to be made.
- BHD Chapter (??): Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- BHD Chapter (1): Still to apply Ken and Bryan's comments.
- SRM beam parameter Chapter (??): Still to apply Ken and Bryan's comments
- MZ chapter Chapter (??): Not written, vaguely planned
- Photodiode Chapter (??): Need to finish adding references.
- Conclusion Chapter (??): Not started. Will write last/almost last

Chapter 1

Design of the LIGO A+ HAM6

Layout

The LIGO A+ HAM6 vacuum chamber contains the readout photodiodes, the output mode cleaners (OMC) and their associated steering optics. Fundamentally, the optical layout in HAM6 must combine the signal beam and the LO beam and sense the combined beams to form a balanced homodyne detector (BHD). The implementation of BHD for LIGO A+ requires a complete redesign of the HAM6 layout as described in this chapter. Section 1.2 is a description of the functionality and design constraints that all candidate HAM6 layouts must satisfy. In Section 1.1, the optical properties of the proposed HAM6 layout are examined. Section 1.3 is an analysis of how mode mismatches between the interferometer and the OMC can be corrected by the Active Wavefront Control (AWC). Section 1.4 is the conclusion for this chapter. This work was undertaken by a team made up of myself, Ken Strain, Russell Jones, Stephen Webster, Mark Barton and Eric Oelker.

1.1 The Chosen HAM6 Optical Layout

The design that is being taken forward at the time of writing is shown in Figure 1.1. Optically, there are many layouts which could have satisfied the requirements and constraints laid out in Section 1.2 . The choice of layout for HAM6 was ultimately determined by mechanical factors such as table space, number of suspensions and preference of Balanced Homodyne Single Suspension (BHSS) design. The optical properties of this layout are summarised in Table 1.1. The mirrors OMA1, OMB1, OMA2, OMB2 etc. will be referred to as OMx1 and OMx2 as, usually, it is not important to make a distinction between the A and B paths.

The beam size and Gouy phase as a function of position along the optical axis is shown in Figure 1.2. Significant points along the optical axis are also shown in this figure. The size of the beam at the final mode-matching optic matches the size of the target mode, as the size of the beam cannot be changed at a mirrors surface. The beam is always less than 1.2 mm within HAM6, an important fact to remember when considering how close structures can be to the beam path.

The astigmatism loss was calculated to be $< 0.05\%$; for detail in this type of calculation, see Appendix B. This astigmatism loss is plotted as a function of angle in Figure 1.4; above 10° the loss becomes significant. The compromise that is made when choosing a larger AOI is between the potential for larger Gouy phase separation between OMx1 and OMx2 at the expense of loss due to astigmatism. In reality the IFO and OMC will also have a small amount astigmatism so this calculation will not be entirely correct, however this effect is negligible compared to other losses. Losses of $< 0.05\%$ are considered unimportant, especially when compared to the

other expected losses, some of which are expected to be at the many percent level, in the readout of the interferometer.

The focal lengths of the OFI lens, OMx1 and OMx2 were optimised so that there was as close to 45° Gouy phase separation between OMx1 and OMx2 as possible. This is shown in Figure 1.3. With the layout shown in Figure 1.1, it is possible to achieve 43° Gouy phase separation between OMx1 and OMx2. Other layouts could provide exactly 45° Gouy phase separation; however, this layout offers many practical benefits, mainly in relation to the BHSS, over the other considered designs that outweigh an additional 2° of Gouy phase separation between OMx1 and OMx2. This layout also provides good Gouy phase separation for alignment control, with there being 88° between OMxS and OMx2.

In conclusion, the proposed HAM6 layout fulfils the criteria set in Section 1.2. It is shown in Figure 1.1. Near optimal mode-matching and alignment control can be achieved with this layout. A summary of the properties of the layout can be found in Table 1.1. The next stage of development for this layout involves the design and prototyping of parts, and the consideration of fine details such as cable routing.

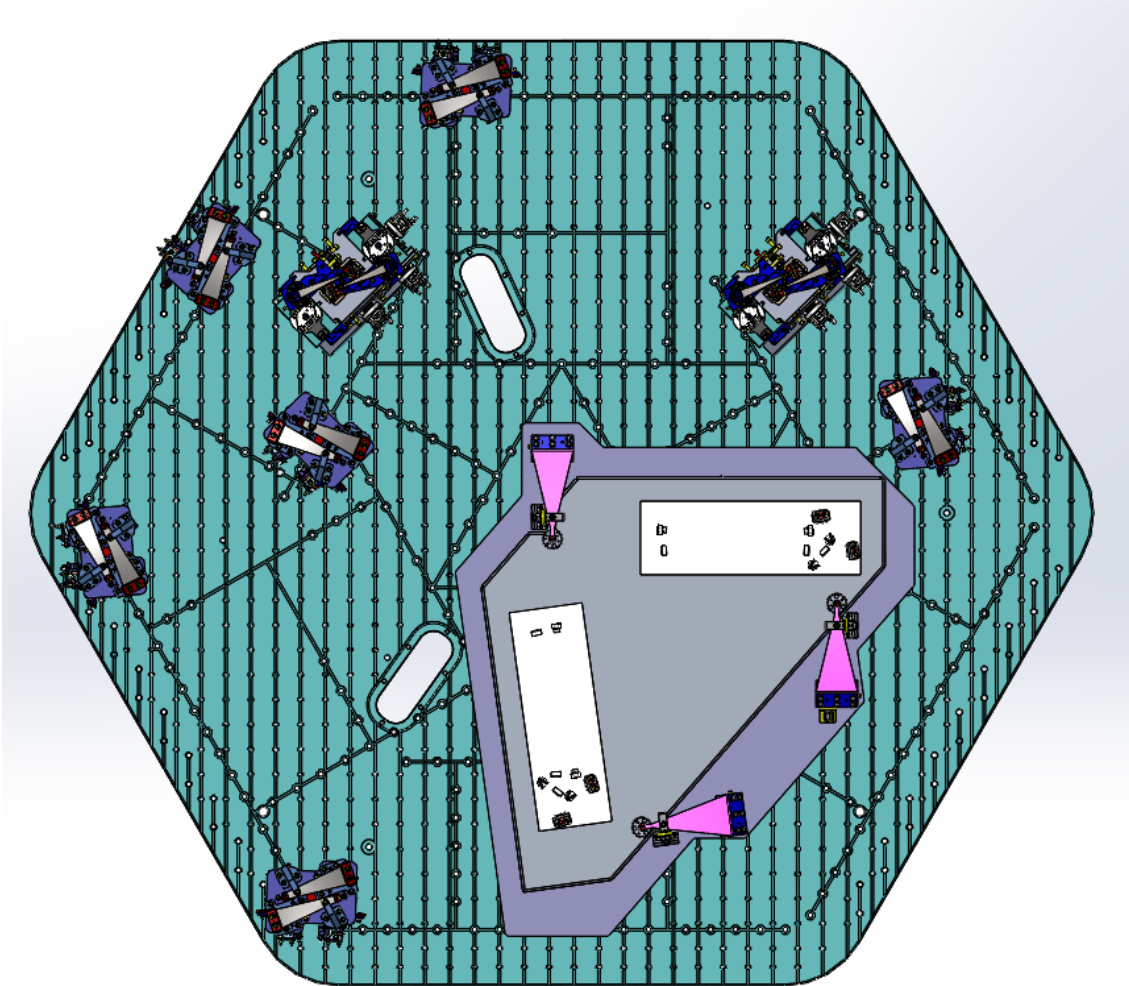


Figure 1.1: Place holder graphic. The pentagram layout. CAD with beam path will be inserted later. Caption will note the main beam path, the bits of hardware mentioned in the text, and will point out that there are lots of other doodads and thingamabobs all over the place which make it tricky to fit everything in HAM6.

Optic	Power (1/m)	Focal length (m)	RoC (m)
OFI lens	-0.066	-15.068	-
OMx1	-0.350	-2.855	5.709 (concave)
OMx2	-0.847	-1.180	2.360 (concave)

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	500
OM0 to BHDBS2	946
BHDBS2 to OMxS	Path A: 225, Path B: 315
OMxS to OMx1	Path A: 1240, Path B: 1150
OMx1 to OMx2	1590
OMx2 to OMCx waist	1220

Parameter	Value
SRM Complex Beam Parameter	$q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$
OMC waist	$485 \mu\text{m}$
OMx1 to OMx2 Gouy phase	42°
OMxS to OMx2 Gouy phase	Path A: 85° , Path B: 81°
OMx1 and OMx2 AOI	A: 9.7° , B: 4.9°

Table 1.1: Mechanical and optical properties of the LIGO A+ HAM6 layout.

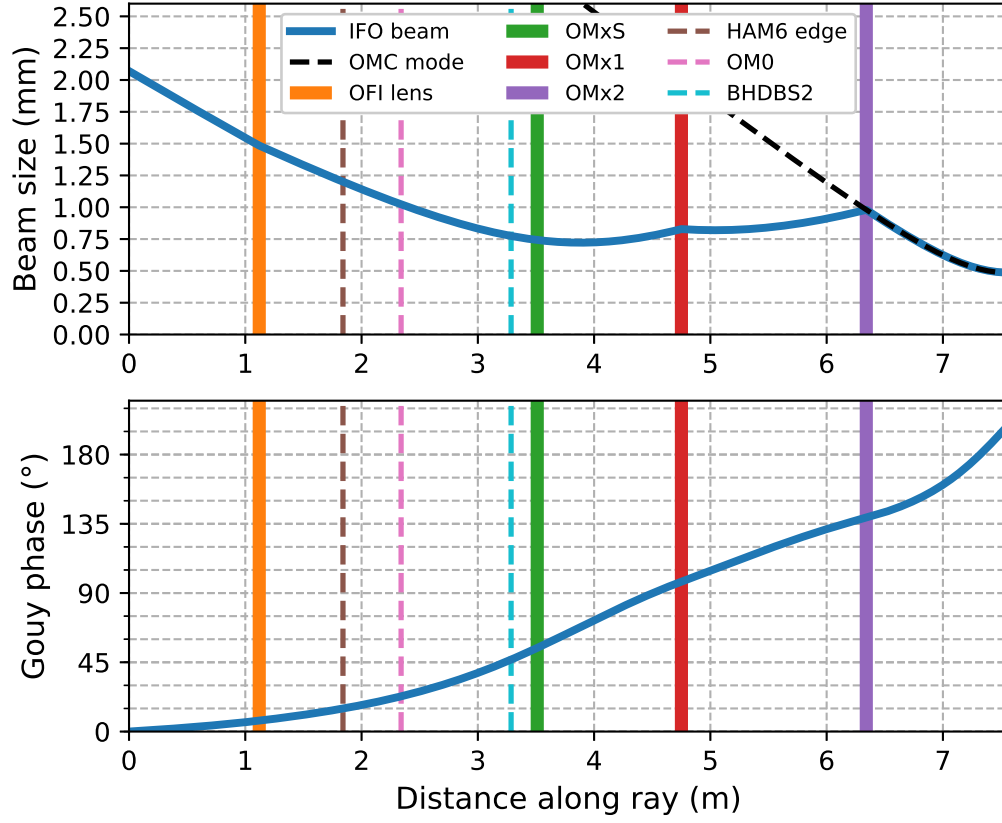


Figure 1.2: Top panel: the radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC. Significant points along the axis are marked with vertical lines. Less significant, but useful, reference points are marked by dashed vertical lines. Bottom panel: The Gouy phase shift accumulated by the beam as it propagates along the optical axis.

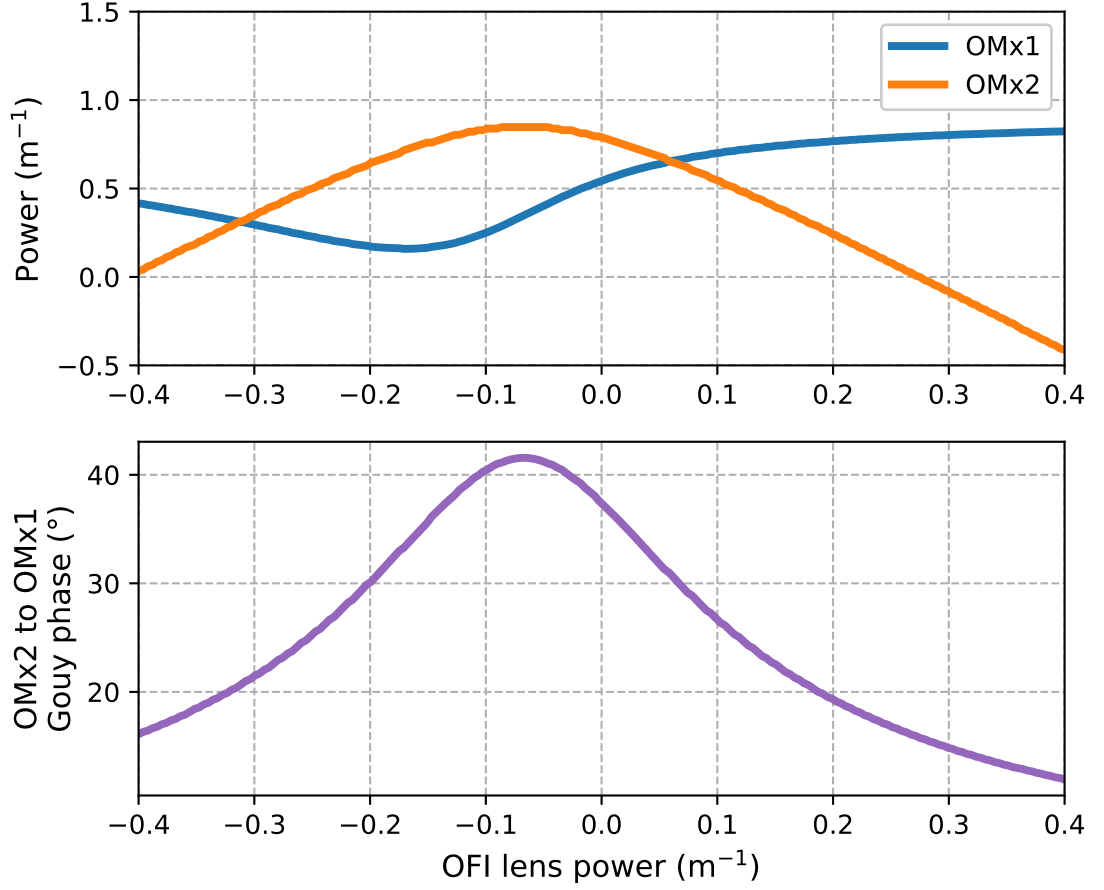


Figure 1.3: Top panel: The choice of OFI lens power affects the combination of OMx1 and OMx2 power that is required for 100% mode-matching. Bottom panel: The best choice of OFI lens corresponds to where the Gouy phase separation between the pair of OMx1 and OMx2 is closest to 45° . The most Gouy phase separation between OMx1 and OMx2 achievable by this layout is 43° .

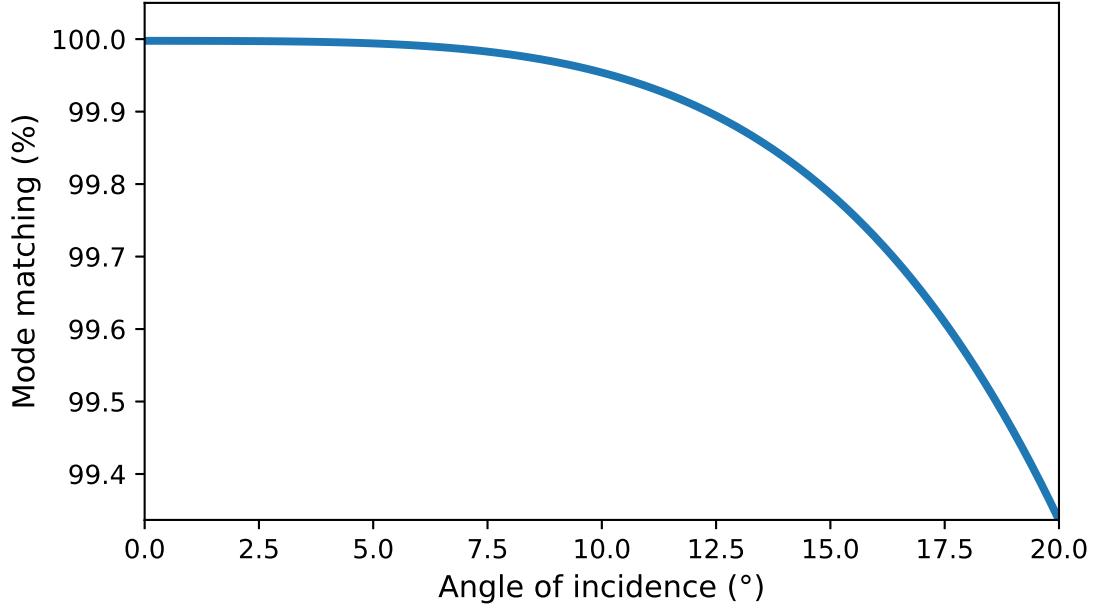


Figure 1.4: Astigmatism is a source of loss. When designing this layout, to explore how sensitive it was to astigmatism, the AOI for the curved mirrors was altered. Using equation (1.2), the loss was computed. This effect is negligible up to 10° after which it starts to become significant compared to the losses summarised in Table 1.2.

1.2 The Requirements for the Optical Layout Within HAM6

HAM6 contains the readout photodiodes which sense the gravitational wave signal encoded on the light. HAM6 must therefore house an optical layout that combines the LO and signal beams as well as providing steering and mode-matching between the interferometer and the OMC ¹. The main design challenge was having control over the two degrees of freedom associated with mode matching – the size and location of the beam’s waist – between the OMC and the beam exiting the SRM as this requires a lot of space relative to the size of the optical table. In addition to this, the HAM6 layout needs to leave space for the other sensors and beams used for controlling the interferometer. This means that all designs used most of the table space.

The goal for mode mismatch between the arm mode and the OMC mode is $< 2\%$; the losses due to the other components in HAM6 is summarised in Table 1.2 [1]. Active mirrors, called Suspended Active Matching Stages (SAMS), will be used to correct for mode-mismatch; they will provide at least ± 50 mD of optical power tuning range [2]. Astigmatism degrades the mode-matching, and non-zero angles of incidence (AOI) of beams onto curved optics cause the reflected beam to become astigmatic (for a comprehensive description of astigmatism see [3]). This prevents one from using a large AOI to increase the separation between optics in HAM6.

¹The OMC assembly includes the readout photodiodes.

The primary role of AWC in HAM6 is to correct for thermal effects in other parts of the interferometer [4] such that the mode-matching between the arm mode and the OMC is maximised. Additionally, the AWC in HAM6 will correct for geometry errors in the interferometer after the main beamsplitter.

Thermal changes in the interferometer’s mode may be corrected for by the Thermal Compensation System (TCS) [5]. However, in practice operators and commissioners use each tunable aspect of the interferometer to improve its sensitivity to gravitational waves. For instance, at LLO it has been found that TCS can be used to decrease laser intensity noise coupling to DARM [6]. As each interferometer subsystem may not be used for its intended purpose, having redundancy in the design of the interferometer is desirable if unexpected noise couplings are discovered. As TCS is used to decrease a noise coupling, it may not be able to correct thermally-driven mode errors. The optical layout and AWC functionality in HAM6 were designed with this flexibility in mind.

While in principle only two curved mirrors in each path to the OMC would be required to mode-match the beam from the interferometer to the OMC – even a single mirror may be able to do that – a lens in the OFI will be used to provide further Gouy phase separation between the two active mode-matching mirrors in each path after BHDBS2. An active lens was considered here, but concerns of OFI assembly heating [7] and the unavailability of suitable flint glass meant that this idea was discarded. Additionally, the range of optical power for the SAMS is large enough that it would be an over-design to implement an active lens in the OFI.

To provide orthogonal mode-matching control, the two SAMS should be separated by 45° of Gouy phase. As most of the Gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will have to travel a significant distance across the table before it acquires sufficient Gouy phase. This Gouy phase separation requirement gave us a method of generating candidate designs as this requirement can be interpreted as needing the mode-matching mirrors to be as far apart as possible from each other on the HAM table. Due to the design of the OMC, the Rayleigh range of the beams is on the order of one meter.

Orthogonal alignment control is obtained by having two mirrors separated by 90° of Gouy Phase. For all designs, the two mirrors for alignment were always immediately after BHDBS2 and immediately before the OMC as this gave a $\sim 90^\circ$ degrees of Gouy Phase separation between them. It was found that placing the first mirror close to BHDBS2 often provided good layouts. Reference [8] describes how the longitudinal and pointing degrees of freedom associated with BHD may be sensed and controlled.

The OMCs are expected to be the largest contributor of back-scatter from HAM6 to the interferometer [4, 9]; As described in [9], backscatter can degrade the sensitivity of the interferometer. A single stage suspended platform (BHSS) is required to prevent backscatter noise from reducing the interferometer's sensitivity. Both OMCs will need to be on this platform and so all designs will need to have the OMCs fairly close to each other. Too large an BHSS is mechanically infeasible as it will reduce space for the other components on the HAM6 table.

Sensors in HAM6 are required to control the alignment between the local oscillator and signal beam, the differential hard mode [?], and the alignment between the squeezer and signal beam. These are shown in Figure ???. The alignment between the signal and local oscillator beams will be monitored using beams which are picked off from the main beam path and so additional mirrors will be required to steer these beams around the chamber. A set of smaller optics will be installed to provide diagnostic beams which can be imaged with CCD cameras. A shutter will be required to protect the hardware in HAM6 from being damaged by the large amounts of light that can potentially be transmitted through the SRM.

1.2.1 Constraints

There are a few properties of the HAM6 layout that cannot be changed. These include: the entry point of beams into the HAM6 chamber, the size of the nominal mode entering HAM6 and the size of the OMC waist². Table extensions can be made in places where they would not impede access, however the size of the HAM optical table does restrict the design. It was decided that the two beam paths after BHDBS2, the optic on which the LO and signal is combined, would be as similar as possible as both beams need to be matched to OMCs of the same design.

The nominal beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$ [10], so the design of the HAM6 mode-matching optics is based on this quantity. As discussed in Chapter ??, the SRM beam parameter is expected to differ from its nominal value. It is described in Chapter ?? how geometry errors in the SRM cause the arm cavity

²In principle they could be changed; however, they will not be as it would require the redesign of large amounts of the interferometer

and the SRC to be mismatched. This means that the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM, and the beam parameter at the SRM that should be considered corresponds to the arm cavity mode. This set of likely modes, shown in Figure ??, was taken into consideration when choosing a layout for HAM6 since different candidate layouts may have been more or less robust to beam parameter errors at the SRM.

The current OMC design developed for aLIGO [9], with trivial modifications, will be used to remove junk light from the signal beam, so there is no choice in target waist size. The nominal OMC waist size is $485\text{ }\mu\text{m}$. The beam enters the OMC through a flat mirror, and a waist is located 140 mm from this mirror. There is a negligible amount of astigmatism inherent in the design of the OMC. Mode-matching optics therefore have to match the nominal SRM and OMC modes.

HAM6 loss Source	Target for O5
OMC throughput loss	$< 2\%$
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	$< 2\%$ (as low as possible)
Any other optical loss	$< 1\%$

Table 1.2: Target for losses in HAM6. Reproduction of [1]

1.3 Visitations of the Robustness of the A+ HAM6 Layout to Mode Mismatch

It is expected that there will be some mismatch between the beam from the interferometer and the OMC due to geometry errors within the interferometer. Additionally,

the thermal state of the interferometer may affect the arm mode beam parameter at the SRM. To visualise the effectiveness of the mode-matching control provided by the layout's active optics, it is useful to consider a 2D space representing Gaussian beams. The region of this space that can be mode-matched by the layout's mode-matching optics and a distribution of modes that those optics may have to correct for can be compared to gain insight into the effectiveness of the layout. It was found that the layout's active mode-matching optics will cover the vast majority of expected input beams.

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, a combination of misalignment and differing beam shape can be considered to be a mode-mismatch. However, mode-matching typically refers to the overlap of the two modes if they were co-aligned, thus it is a measure of how similar in shape the two modes are.

1.3.1 A Description of Mode-Matching in Terms of HG Modes

Mode-matching is usually, to first approximation, thought of as a difference in the beam's waist size and the beam's waist location. First order mode-mismatch can be modelled as light scattering from the beam's, or cavity's, fundamental mode into its $\text{HG}_{20} + \text{HG}_{02}$ mode. If a cavity is designed such that the fundamental mode will be resonant while the $\text{HG}_{20} + \text{HG}_{02}$ mode will be reflected then the effect of mode-mismatch between the cavity and a signal carrying beam is to decrease the signal light that is transmitted by the cavity; since, some of the signal carrying light will be scattered into the reflected $\text{HG}_{20} + \text{HG}_{02}$ mode. Sensing the $\text{HG}_{20} + \text{HG}_{02}$ mode therefore gives a measure of the mode-matching between two lasers.

A change in waist size is equivalent to an in-phase coupling between the fundamental mode and the $\text{HG}_{20} + \text{HG}_{02}$, whereas a change in waist location is a coupling to the $\text{HG}_{20} + \text{HG}_{02}$ which is 90° out of phase with the fundamental mode [11]. As the $\text{HG}_{20} + \text{HG}_{02}$ acquires 2ϕ compared to the fundamental mode, where ϕ is the Gouy phase acquired by the fundamental mode, spacing beam curvature actuators at 45° will give an orthogonal basis for control over mode-matching. This means that the beam size and beam waist, or two combinations of them that are orthogonal, can be independently controlled.

1.3.2 Calculating Mode-Matching Using The Gaussian Beam Parameter

Two perfectly mode-matched beams will have identical beam parameters at the same point along the optical axis. The overlap integral between two fields, E_1 and E_2 ,

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}, \quad (1.1)$$

can be used to determine the amount of mode-matching, $\mathcal{M}$, between two astigmatic modes. Solving this integral gives

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}, \quad (1.2)$$

where $q_{1,2}$ is the beam parameter of the electric field $E_{1,2}$. This is derived in Appendix B. The solution to equation (1.2) is always $0 < \mathcal{M} \leq 1$. Alternatively, mode-matching may be computed in terms of the amplitude of the higher order modes relative to the fundamental mode [12]. The mode-matching can be calcu-

lated at any point along the optical axis of the layout as mode-matching must be preserved throughout it; however, it makes sense to calculate the mode-matching in a physically significant plane such as the input plane, a plane containing a curved optic or the plane containing the waist of the target beam.

1.3.3 WS Space Visualisations

A Gaussian beam can be described by its beam size, w , and defocus, S [13]. A WS space plots a Gaussian beam in terms of these two parameters. The effect of focussing elements on Gaussian beams can be clearly seen in this representation of them. Because a curved mirror can only change the curvature of a beam, at the mirrors surface the beams before and after reflection must have the same size. The process of a Gaussian beam, described by $q_1(S_1, w_1)$, being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i\frac{\lambda}{\pi w_1^2}. \quad (1.3)$$

From Equation (1.3) it can be seen that a mode moves in the S direction when it reflects from a curved mirror. This representation of Gaussian beams has the advantage of showing the beam in terms of intuitive quantities. The downside to this representation is as the beam propagates through the layout, the path that is traced through the WS space by the mode is fairly complex.

There are two other ways that a beam parameter could be graphically represented. Since q is a complex number, it could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but

the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and S produces plots where the effect of focussing elements and propagation are easy to visualise [14, p.680]. Curved mirrors and lenses result in a change in defocus, and propagated modes follow circular paths. As these figures are being used to visualise mode-matching and the effect of actuators on the beam size (rather than $\lambda/\pi w^2$) and defocus, the WS representation was chosen. One could express the beam parameter in terms of natural units by dividing the waist by w_0 and the defocus by $1/z_R$.

To illustrate how actuators move a mode in WS space, consider two lenses with variable power separated by 45° of Gouy phase. As the two lenses are orthogonal, the waist size and waist location of the mode exiting this layout will be orthogonally controllable. In a WS space the orthogonality of two actuators can be seen by the angle between the two vectors that represent each actuation. If the two lenses are orthogonal then the actuation vectors will be orthogonal. Figure 1.5 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase.

The mode-matching requirement between the interferometer and the OMC for LIGO A+ is $> 98\%$. It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode; the mode-matching between two arbitrary modes can not be deduced from these contours [13]. For example two modes may be mismatched by the same amount from the target in orthogonal directions in WS space. They cannot be 100% mode-matched as these

two modes are different from each other.

1.3.4 Visualising how Active Mode-Matching Optics Expand the Area of WS Space Representing Acceptable Input Modes

To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS space at the SRM from which the target mode-matching space can be achieved was computed. To do this, first the ring of modes in WS space that is 98% mode-matched to the OMC after reflection from the second active mirror, OMx2, was considered. If OMx2 had a static radius of curvature then the effect of OMx2 would be to move the ring vertically in the WS plane. However, the ring can also be stretched in the S direction as OMx2's radius of curvature can be changed. This can be seen, crudely, by applying $S_{\text{OMx2}} \pm \Delta S$ to the original $> 98\%$ mode-matching ring to produce two new rings separated by $2\Delta S$.

To generate the region from which the target region can be reached by OMx2, the target region should be cut by a line which intersects the ring at the extrema of the beam size. To maximise the amount of WS space covered by the actuator, the upper target region should have $+\Delta S$ applied to it, whereas for the lower region the actuation should be $-\Delta S$. This results in a region of WS space which is larger than the initial target region. All modes in the new region can be at least 98% mode-matched to the OMC. The new region was then propagated to the first mirror, OMx1, where the same procedure can be applied to further expand the region. This

process is shown in Figure 1.6.

The modes from the SRM that can be matched to the target is shown in Figure 1.7. To visualise how successful the layout is at handling the likely modes from the interferometer (this set of modes was investigated in Chapter ??), the interferometer modes were compared to the region of WS space containing modes which can be corrected such that they meet the $> 98\%$ mode-matching requirement. It was found that almost all likely modes will be covered by the layout, as the modes near the edge of the interferometer mode continuum are more unlikely to occur than the ones near the centre. Its worth noting that the in Figure 1.7, modes at the yellow end of the spectrum may not be physical as they correspond to an unstable SRC.

For all candidate layouts analysed there was no major difference in the amount of modes that could be mode-matched. In principle, the requirement for having 45° of Gouy Phase separation between OMx1 and OMx2 does not maximise the amount of *likely* interferometer modes that can be mode-matched. It is be possible to shape the region of WS space covered by the layout to maximise the probability that the arm mode will be at least 98% mode-matched to the OMC by the layout. For example, if it was found that arm modes were most likely to be characterised by an error in beam waist size at the OMC, the active optics should be arranged so that they can actuate predominantly on beam size at the OMC. However, choosing a layout which favours flexibility in one degree of freedom over the other may lead to unforeseen beam parameter errors being uncorrectable.

The shape of the pink region in Figure 1.7 is nominal; when the HAM6 layout is constructed the pink regions shape will be altered by geometric errors in OMx1 and

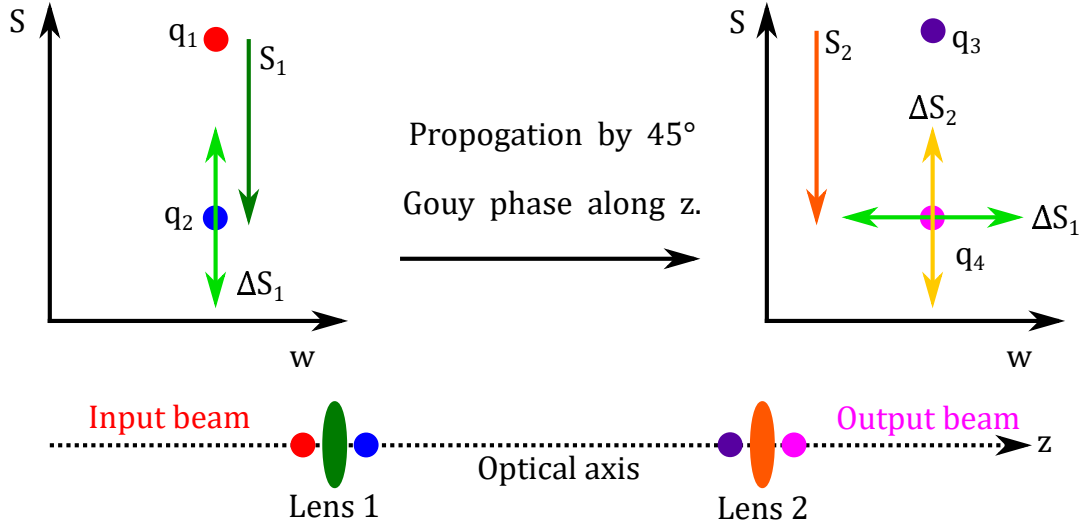


Figure 1.5: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} \pm \Delta S_{1,2}$. The left hand axis shows the WS plane at Lens 1. The input beam q_1 is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 which is shown as a blue dot. A light green arrow indicates the actuation of Lens 1. The right hand axis shows the WS plane at Lens 2. The input mode at Lens 2, q_3 , is shown as a purple dot, and the output mode, q_4 , is shown as a pink dot. The static and dynamic power of Lens 2 is indicated by the orange and yellow lines. As Lens 1 and Lens 2 are separated by 45° of Gouy Phase, the actuation provided by Lens 1 results in a change in the beam parameter which is orthogonal to the actuation provided by Lens 2, provided that the actuations are small enough. Beneath the two WS space diagrams is a sketch of the optical axis and where the modes are along it.

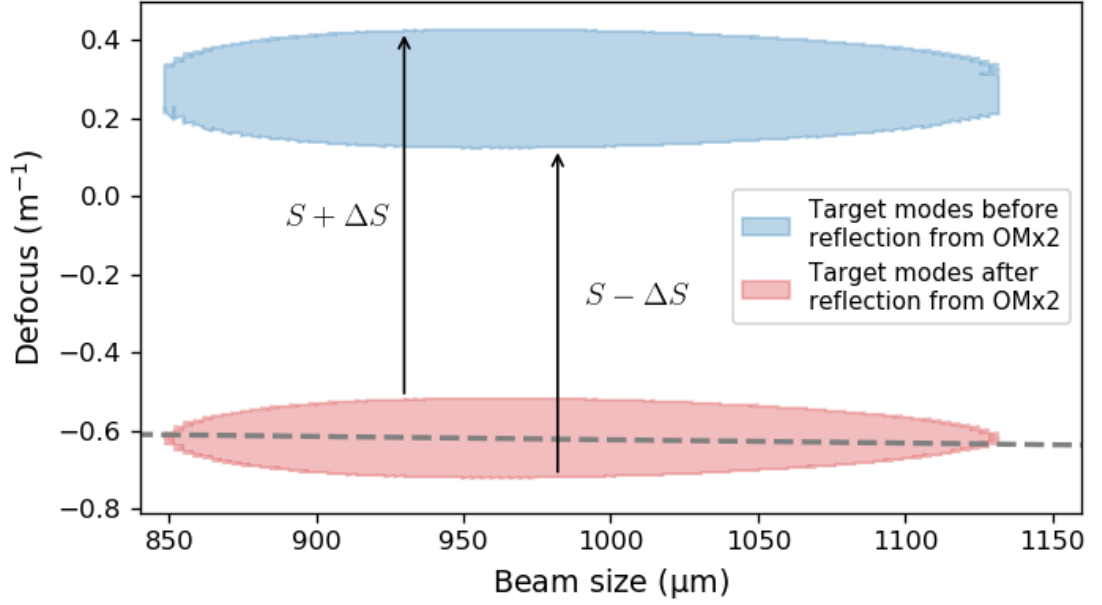


Figure 1.6: This is the WS space at the OMx2 plane. The effect of OMx2 in the plane of OMx2 is to move a mode vertically in WS space. After reflection from OMx2, a mode must be at least 98% mode-matched to the OMC as there are no further active mode-matching elements in the layout of HAM6. Modes with at least 98% are shown by the red region. The blue region represents which modes can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, indicated by the dashed line that connects the two extremes in width of the region. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, the upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them.

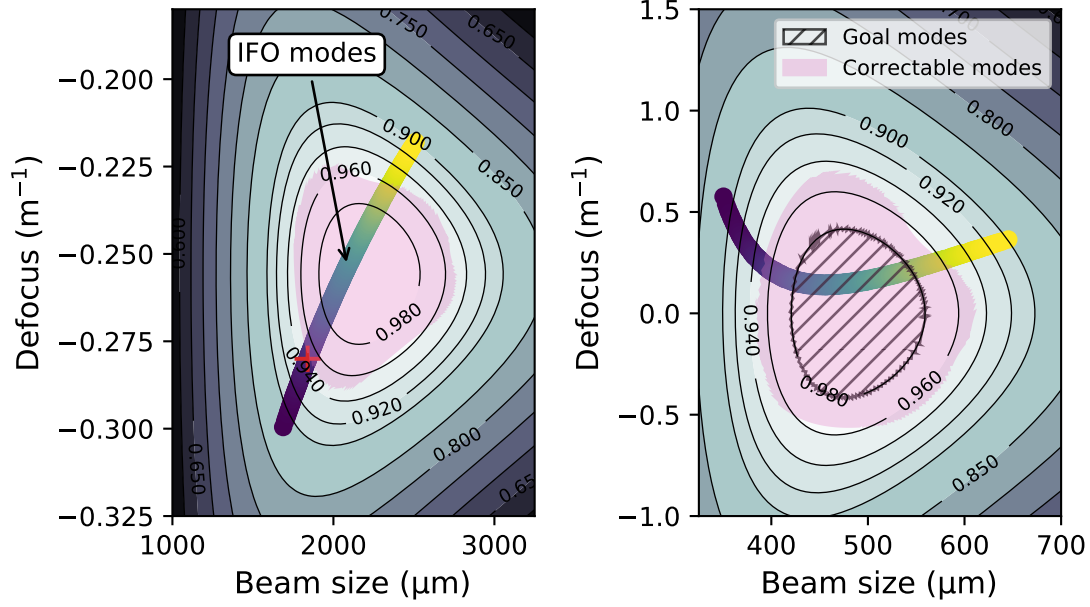


Figure 1.7: The region of WS space covered by the AWC in HAM6 if OMx1 and OMx2 are separated by 45° of Gouy phase. Left panel: the set of modes that could be present at the SRM, as described in Chapter ??, is shown by the purple-blue-green-yellow curve. The pink region shows which modes can reach 98% mode-matching by using the AWC in HAM6. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. Right panel: this shows the same modes as the left panel, except they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region, which is indicated with a hatched area.

OMx2. Nevertheless, Figure 1.7 is encouraging since it shows that the actuation provided by OMx1 and OMx2 is highly tolerant to mode-matching errors generated by errors in the SRC.

1.3.5 Optimising WS Space Coverage With a Super Actuator

While 45° of Gouy phase separation between the mode matching optics in HAM6 is a desirable quality for it to have, with an actuation range ± 50 mD, the SAMS may not be able to correct for the likely set of modes that could exit the interferometer. This can be seen in Figure 1.7. Sacrificing the orthogonality of the mode matching optics can help correct for expected beam parameter errors.

The majority of the uncertainty in the interferometer's beam parameter comes from the uncertainty in SR3's RoC (see Figures ?? and ??). Ideally, one would correct for this by actuating on the RoC of SR3, however the RoC of SR3 cannot be strongly actuated on. If a mirror was placed 180° away from SR3, it could be used to compensate for the uncertainty in the RoC of SR3 i.e. one could configure the mode matching optics within HAM6 such that they are biased for correcting the errors in the interferometer beam parameter caused by the error in SR3.

A super actuator is comprised of two or more actuators that act on the same property. In this way, the range of errors in that can be corrected for in this property is increased compared to if only one actuator is used. The ideal Gouy phase separation for a super actuator would be close to 0° , however low amounts of Gouy phase separation ($\sim 30^\circ$) offer a compromise between orthogonality versus the ideal super actuator.

Figure 1.8 shows how the region of WS space covered by the HAM6 AWC can be biased in favour of the expected set of modes that may emerge from the SRM. The physical positions of the optics are the same as before, however the optical powers of the OFI lens, OMx1 and OMx2 was altered (-0.2, 1.67, 0.631 respectively) to change the Gouy phase separation between OMx1 and OMx2 to 30°.

1.4 Conclusion

The functionality required from the BHD layout was discussed, and a proposed layout that satisfies these requirements is shown in Figure 1.1. The optical properties of this layout were investigated. By selection of the right combination of mode-matching optics, near orthogonal control of the signal-LO mode-matching and beam alignment to the OMC was achieved.

The effect that the AWC had on the region of WS space that contains modes which satisfy the $> 98\%$ mode-matching requirement space was investigated. WS diagrams were used to represent how robust the layout is to initial mode-mismatch. Figure 1.7 shows that the AWC provides a huge amount of mode-matching range. This means that the layout can tolerate large changes in the beam parameter at the SRM.

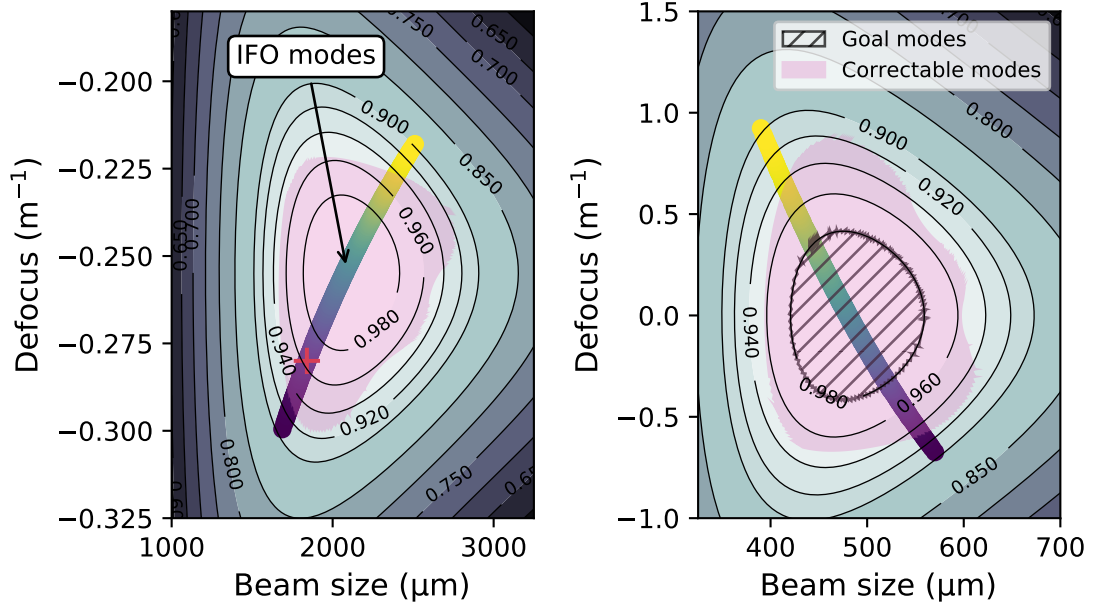


Figure 1.8: At the expense of orthogonality, the region of WS space covered by the AWC in HAM6 can be optimised by changing the Gouy Phase separation between OMx1 and OMx2. Left panel: the set of modes that could be present at the SRM, as described in Chapter ??, is shown by the purple-blue-green-yellow curve. The pink region shows which modes can reach 98% mode-matching by using the AWC in HAM6. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. Right panel: this shows the same modes as the left panel, except they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region, which is indicated with a hatched area.

Appendix A

Algorithm For Finding a Three Optic Solution with 45° Gouy Phase

Separation between two of the Optics

To find a mode matching solutions between the SRM and the OMCs which have 45° phase separation for the AWC optics, an algorithm was developed. To visualise how this algorithm works see figures A.2 and A.1. Since a beam can be described by two parameters, its defocus S and width w , the modes can be shown in a 2-D space [13] (figure A.2). The action of a lens or curved mirror is to move the mode in the S direction. It is helpful to define a mode continuum along which w and/or S varies. The principle by which this algorithm works is: The continuum of modes after optic 1 and the continuum of modes just before optic 3 can be easily obtained since they must have a defined beam size, so they only vary in defocus. These two continuum are propagated to optic 2, and then its just a case of finding how to move from one continuum to an other.

The continua at optic 2 do not necessarily have to have the shape that they are sketched with in figure A.2, however if the blue-green and red-yellow continuum extend far enough they will always wrap around. It is also not guaranteed that every mode in the red-yellow can be reached from the blue-green continuum, and vice versa. Since the mapping between the blue-green and red-yellow continuum is not one to one, there is some choice in you can make. Both ‘branches’ of the continua, where the continuum turns back on itself in the w direction, do not need to be analysed at the same time, but a better solution may be found by looking at different combinations of branches. This is useful in setting a limit on the size of the continuum you search.

In our implementation of this algorithm, the continua are actually discrete, which means that when finding a mapping at optic two it is not guaranteed that a mode in the red-yellow continuum will have a corresponding mode of the same with in the blue-green continuum. Because of this granularity, if you were to plot optimal values of optic 2 and optic 3 power as a function of optic 1 power, you would see jumps in the optimal lens choices. This is especially bad around where the turning point in the continua is, since the modes defocus changes quickly while their size does not.

A.0.1 Method

This section delineates the method described in section A and can be visualised using figure A.2.

Define the target mode at the optic 3. In figure A.2, this is mode l at optic three.

Define a set of modes which share the same beam size as the target mode at optic 3 but have varying radii of curvature. In figure A.2, this is the red-yellow continuum at optic three.

Propagate the set of modes defined at optic 3 back to the optic 2. In figure A.2, this is the red-yellow continuum at optic 2.

Define the initial beam parameter and propagate this to optic 1. In figure A.2, this is mode i at optic one.

Define a set of modes that have the same beam size as the initial beam at optic 1 but differ in defocus. In figure A.2, this is the blue-green continuum at optic one.

Propagate the modes defined at the optic 1 to optic 2. In figure A.2, this is the blue-green continuum at optic two.

Calculate the difference in optical power between mode i and a mode in the blue-green continuum (e.g. mode j). In figure A.2, mode j represents just one possible mode in this continuum. Save this power.

Find a mode in the red-yellow continuum which is closest in size to mode j, call this mode k. In figure A.2, there is another possible choice on the red-yellow continua, however only one of them needs to be considered.

Calculate the difference in optical power between mode j and mode k at optic 2. Save this power.

Calculate the difference in optical power mode k and the target mode at optic 3. Save this power.

Calculate the Gouy phase separation of optic 2 and optic 3.

Repeat for every mode in the blue-green continuum, or until a gouy phase separation of 45degree has been found.

The number of modes that are considered and the range of modes in the continua should be tailored for the optical layout that is being considered. Our problem was with metre scale spacings between optics, and with metre scale Rayleigh lengths. It was found that continua with a range of the order 1-10 dioptries were suitable. This does not mean that an adaptive optic would need this actuation range, rather it allowed for the static radii of curvature for each optic that gives optimal Gouy phase separation between the active mode matching elements to be found. Using this algorithm, we were able to rapidly develop many possible HAM6 layouts. This allows for more time to consider various engineering factors (suspension size and the like) and how to adapt the mirror strengths accordingly to accommodate for small changes in a tentative design.

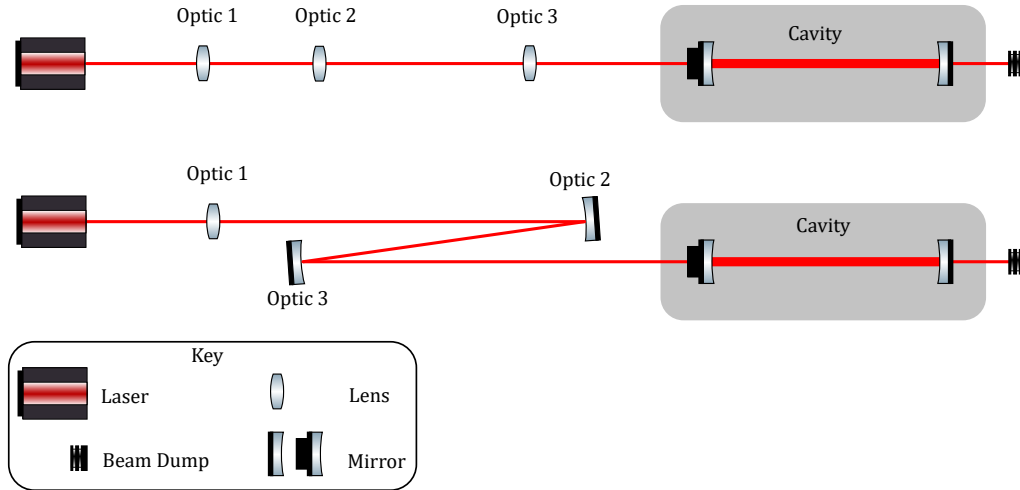


Figure A.1: Top layout: Three lenses used to achieve mode matching to the cavity. These are shown as positive lenses, however negative lenses can also be used when designing mode matching systems. Bottom Layout: A lens and two curved mirrors to achieve mode matching to a cavity. The optics 1,2 and 3 correspond to the ones shown in figure A.2. The initial and target modes are defined in these examples by the laser mode and the cavity mode.

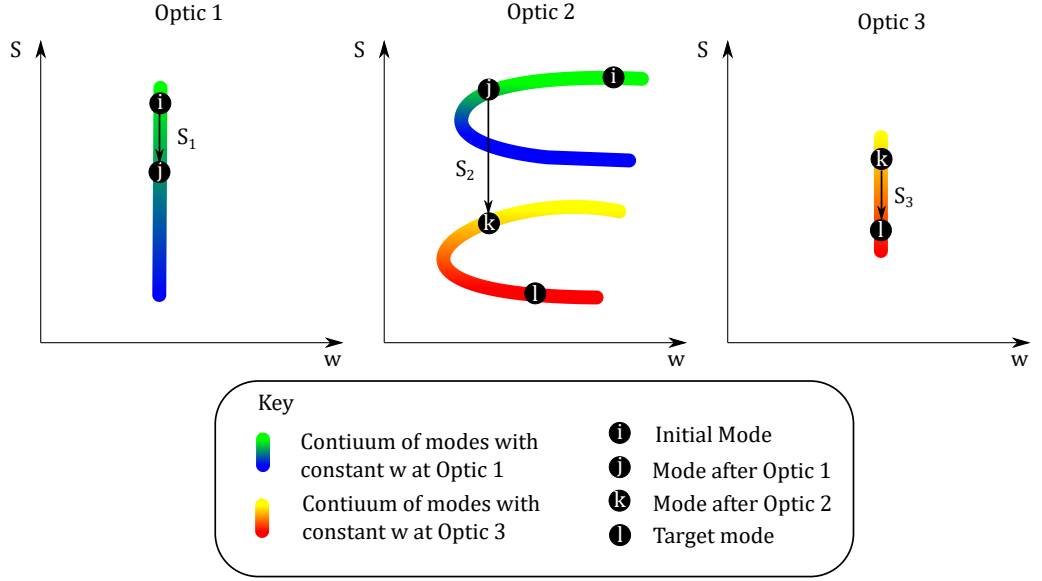


Figure A.2: Sketch of the WS phase space representations of a Gaussian beams at the three positions along the optical axis where a mirror or a lens could be. Examples of such layouts are shown in figure A.1. The optical power of the first optic is given by the difference in the defocus between the initial mode (i) and the second mode (j). The power of the second optic is found by finding the difference in defocus from the blue-green continuum to the red-yellow continuum. There are 2 optical powers that will move the mode j to the red-yellow continuum, and two possible modes in the blue-green continuum that can from which k can be reached, giving 4 possible similar solutions. The algorithm can be set up so that it is only one way of moving between the blue-green and red-yellow continua by limiting their extent. The power of the third optic is found from the difference between the mode k and mode l at optic 3.

Appendix B

Mode Matching Between 2 Astigmatic Beams

B.1 Derivation of Mode Matching TM00 Beams

The [check this naming](#) electric field, E , of a TM00 Gaussian beam can be written as

$$E(x, y, z) = E_x(x, z)E_y(y, z) \quad (\text{B.1})$$

where

$$E_x(x, z) = \frac{1}{\sqrt{q_x(z)}} \exp\left(-ik \frac{x^2}{2q_x(z)}\right), \quad (\text{B.2})$$

$$E_y(y, z) = \frac{1}{\sqrt{q_y(z)}} \exp\left(-ik \frac{y^2}{2q_y(z)}\right). \quad (\text{B.3})$$

For a beam without astigmatism, $x^2 + y^2 = r^2$ and $q_x = q_y = q$ so:

$$E(r, z) = \frac{1}{q(z)} \exp\left(-ik \frac{r^2}{2q(z)}\right). \quad (\text{B.4})$$

The mode matching, $\mathcal{M}$, of two beams can be expressed by overlap of the two fields

$$\mathcal{M} = \frac{\left| \int_A E_1^* E_2 dA \right|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}. \quad (\text{B.5})$$

The numerator of equation B.5 for TM00 beams can be written as:

$$\left| \int_A E_1^* E_2 dA \right| = \int_A E_1^* E_2 dA \int_A E_1 E_2^* dA \quad (\text{B.6})$$

$$= \int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{y1}^* E_{y2} dy \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx \int_{-\infty}^{\infty} E_{y1} E_{y2}^* dy. \quad (\text{B.7})$$

Since the calculation is the same for the x and y coordinates, we only need to show the solution for the x . The x terms of equation B.7 can be written as:

$$\int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}^*}} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \frac{1}{\sqrt{q_{x2}}} \exp\left(\frac{-ikx^2}{2q_{x2}}\right) dx \quad (\text{B.8})$$

$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}}} \exp\left(\frac{-ikx^2}{2q_{x1}}\right) \frac{1}{\sqrt{q_{x2}^*}} \exp\left(\frac{ikx^2}{2q_{x2}^*}\right) dx \quad (\text{B.9})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \exp\left(\frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{B.10})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*} - \frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{B.11})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right). \quad (\text{B.12})$$

Using the identity

$$\mathcal{I} = \int_{-\infty}^{\infty} \exp (i a x^2) d x = \frac{i \pi}{a} \quad (\text{B.13})$$

and labelling

$$a = \frac{k \left(q_{x2} - q_{x1}^* \right)}{2 q_{x1}^* q_{x2}} \quad (\text{B.14})$$

allows equation B.12 to be simplified to

$$\frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp \left(\frac{i k x^2 (q_{x2} - q_{x1}^*)}{2 q_{x1}^* q_{x2}} \right) d x \times c.c. \right) = \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{i \pi (2 q_{x1}^* q_{x2})}{k (q_{x2} - q_{x1}^*)}} \sqrt{\frac{-i \pi (2 q_{x1} q_{x2}^*)}{k (q_{x2}^* - q_{x1})}} \quad (\text{B.15})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{4 \pi |q_{x1}|^2 |q_{x2}|^2}{k^2 |q_{x2} - q_{x1}^*|^2}} \quad (\text{B.16})$$

$$= \frac{2 \pi}{k} \frac{1}{|q_{x2} - q_{x1}^*|}. \quad (\text{B.17})$$

This means that the numerator of equation B.5 is

$$\frac{4 \pi^2}{k^2} \frac{1}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{B.18})$$

Since E_x and E_y are orthogonal, $|E|^2 = |E_x|^2 |E_y|^2$, an expression for the denominator of equation B.5 can be found by:

$$\int_A |E|^2 dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E_x|^2 |E_y|^2 d x d y \quad (\text{B.19})$$

$$= \int_{-\infty}^{\infty} \frac{1}{|q_x|} \exp \left(\frac{i k x^2 (q_x - q_x^*)}{2 q_x q_x^*} \right) \times y \text{ term.} \quad (\text{B.20})$$

Using a similar approach as in the calculation of the numerator terms:

$$\int_A |E|^2 dA = \frac{1}{|q_x|} \sqrt{\frac{i2\pi |q_x|^2}{k(q_x - q_x^*)}} \frac{1}{|q_y|} \sqrt{\frac{i2\pi |q_y|^2}{k(q_y - q_y^*)}} \quad (\text{B.21})$$

$$= -\frac{2\pi}{k} \frac{1}{\sqrt{(q_x - q_x^*)(q_y - q_y^*)}} \quad (\text{B.22})$$

we get the mode matching $\mathcal{M}$ to be

$$\mathcal{M} = \frac{\frac{4\pi}{k^2}}{-\frac{2\pi}{k} \times -\frac{2\pi}{k}} \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)} \sqrt{(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|} \quad (\text{B.23})$$

$$= \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{B.24})$$

B.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam

The error in the optical power of a mirror due to an error in the radius of curvature of is

$$S = \frac{2}{R_m} \quad (\text{B.25})$$

$$dS = \frac{-2}{R_m^2} dR. \quad (\text{B.26})$$

The error in the beams defocus is the same as the error in the radius of curvature of the the mirror. The effect this has on the beam parameter q can be found by

defining a contour of modes distance dq from q i.e. $q + |dq| e^{i\theta}$, where dq is

$$q = \frac{q_i}{Sq_i + 1} \quad (\text{B.27})$$

$$dq = \frac{q_i^2}{(-Sq_i + 1)^2} dS. \quad (\text{B.28})$$

This gives you a family of modes which will contain the modes due to an error in the radius of curvature of the mirror, but since q is a complex number the contour you need to define to encapsulate the concept of an error in q also contains modes which are have different beam sizes as well as different defocuses. To do the calculation this way is cumbersome, it is much more straight forward to compute the effect of having a mirror ROC error by evaluating at every mirror

$$\begin{pmatrix} 1 & 0 \\ \pm dS & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix} \begin{pmatrix} q \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix} \quad (\text{B.29})$$

and then divide the top and bottom entries of the $\begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix}$ vector.

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